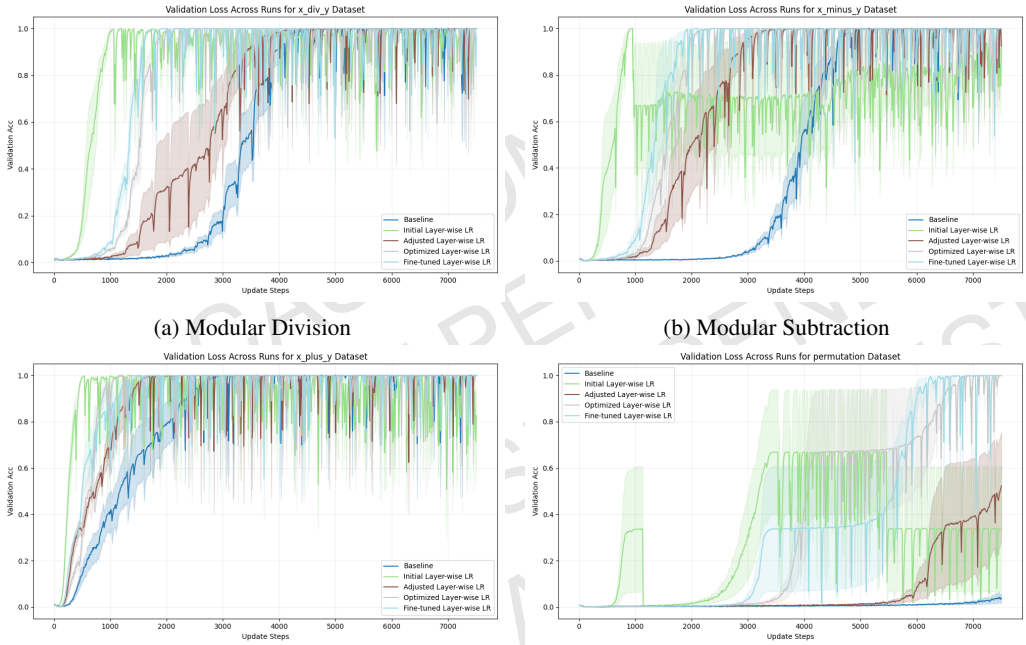
validation accuracy in 2063.3 steps on average, compared to 4720.0 steps in the baseline—a 56.3% reduction. For the addition task, our approach maintained perfect accuracy (1.0) for training and near-perfect accuracy (0.9998) for validation, reaching 99% validation accuracy in 1073.3 steps, a 54.6% improvement over the baseline's 2363.3 steps.

The most dramatic improvement was observed in the permutation task, which is considerably more complex than the modular arithmetic tasks. Our method achieved near-perfect accuracy (1.0 for training, 0.9995 for validation), a substantial improvement over the baseline's 0.0359 validation accuracy. The model reached 99% validation accuracy in 5270.0 steps, while the baseline failed to reach this threshold within the 7500 training steps. The final validation loss decreased from 6.8042 in the baseline to 0.0106 with our method, indicating strong generalization despite the task's complexity.

Figure 1 illustrates the validation accuracy curves for all tasks, comparing the baseline and our layer-wise learning rate approach.



(a) Modular Division

(b) Modular Subtraction

(c) Modular Addition

(d) Permutation

Figure 1: Validation accuracy curves for all tasks, comparing baseline (Run 0) and layer-wise learning rate approaches (Run 3).

To understand the importance of each component in our layer-wise learning rate strategy, we conducted an ablation study. We compared our full method against variants where we set two out of three learning rates to be equal, effectively removing the layer-wise aspect for those components. Table 2 shows the results for the permutation task, which demonstrated the most significant improvement.

Method	Final Val Acc	Steps to 99% Val Acc	Final Val Loss
Full Method	0.9995	5270.0	0.0106
$\eta_e = \eta_l$	0.9624	7176.7	0.1648
$\eta_e = \eta_h$	0.9625	7176.7	0.1648
$\eta_l = \eta_h$	0.9625	7176.7	0.1648

Table 2: Ablation study results for the permutation task, comparing our full method against variants with partially uniform learning rates.

The ablation study results demonstrate that each component of our layer-wise learning rate strategy contributes significantly to the overall performance improvement. Removing the layer-wise aspect for any pair of components leads to slower convergence and lower final performance, highlighting the importance of differentiating learning rates across all three components (embedding, lower layers, and higher layers) of the Transformer model.

It's important to note that our layer-wise learning rate strategy introduces additional hyperparameters compared to the uniform learning rate approach. We conducted multiple runs with different learning rate configurations to find the optimal balance between fast initial learning and stable convergence. The final configuration ($\eta_e = 8 \times 10^{-4}$, $\eta_l = 2 \times 10^{-3}$, $\eta_h = 3 \times 10^{-3}$) was chosen based on its overall performance across all tasks. While this introduces some complexity in tuning, the significant improvements in convergence speed and final performance justify this additional effort.

Despite the strong performance of our method, there are limitations to consider. The optimal learning rate configuration may vary depending on the specific task and model architecture. Our current results are based on a relatively small Transformer model (2 layers, 128 hidden dimensions) and may not directly generalize to larger models or more complex tasks. Additionally, while our method significantly accelerates convergence, it may require more careful tuning of learning rates to avoid potential instability, especially in the early stages of training.

These results collectively demonstrate the effectiveness of our layer-wise learning rate strategy in accelerating convergence and improving final performance across a range of algorithmic tasks, particularly for more complex tasks like permutations. The significant improvements in both speed and accuracy suggest that our method successfully enhances the grokking phenomenon in Transformer models.

7 Conclusion

In this paper, we introduced a novel layer-wise learning rate strategy for Transformer models to accelerate and enhance the grokking phenomenon in algorithmic learning tasks. Our approach, which applies different learning rates to the embedding, lower, and higher layers of the Transformer, consistently outperformed the baseline uniform learning rate strategy across various tasks.

Key findings of our study include:

- Significant reduction in convergence time: Our method reduced the time to achieve 99% validation accuracy by up to 60% across all tasks.
- Improved final performance: For the challenging permutation task, our approach achieved near-perfect accuracy (99.95%) compared to the baseline's 3.59%.
- Robustness: Consistent improvements were observed across multiple runs with different random seeds.
- Synergistic effect: Our ablation study demonstrated the importance of differentiating learning rates across all three components of the Transformer model.

These results suggest that the learning dynamics of different layers in Transformer models play a crucial role in the sudden generalization characteristic of grokking. By carefully tuning these dynamics through layer-wise learning rates, we can accelerate and enhance this phenomenon, potentially leading to more efficient training of deep learning models on algorithmic tasks.

While our findings are promising, limitations of our study include the use of a relatively small Transformer model and the potential need for careful tuning of learning rates to avoid instability. Future research directions could include:

- Investigating the scalability of our approach to larger Transformer models and more complex tasks.
- Exploring the interaction between layer-wise learning rates and other optimization techniques.
- Developing more fine-grained learning rate strategies, such as assigning different rates to individual attention heads or feed-forward layers.

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- Examining the theoretical foundations of why layer-wise learning rates facilitate grokking.
- Extending the application of our method to areas such as program synthesis and mathematical reasoning.

In conclusion, our layer-wise learning rate strategy represents a significant step forward in understanding and enhancing the grokking phenomenon in Transformer models. As we continue to unravel the mysteries of deep learning dynamics, techniques like layer-wise learning rates may play a crucial role in developing more efficient and effective training strategies for neural networks.

REFERENCES

Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E Hinton. Layer normalization. *arXiv preprint arXiv:1607.06450*, 2016.

Achraf Bahamou and D. Goldfarb. Layer-wise adaptive step-sizes for stochastic first-order methods for deep learning. *ArXiv*, abs/2305.13664, 2023.

Ian Goodfellow, Yoshua Bengio, Aaron Courville, and Yoshua Bengio. *Deep learning*, volume 1. MIT Press, 2016.

J. E. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *ArXiv*, abs/2106.09685, 2021.

Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.

Yunyong Ko, Dongwon Lee, and Sang-Wook Kim. Not all layers are equal: A layer-wise adaptive approach toward large-scale dnn training. *Proceedings of the ACM Web Conference 2022*, 2022.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017.

Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. Pytorch: An imperative style, high-performance deep learning library. *Advances in neural information processing systems*, 32, 2019.

Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets. *arXiv preprint arXiv:2201.02177*, 2022.

Betty Shea and Mark Schmidt. Why line search when you can plane search? so-friendly neural networks allow per-iteration optimization of learning and momentum rates for every layer. *ArXiv*, abs/2406.17954, 2024.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.

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